

Data Analysis

Load Space X dataset, from last section.

```
[3]: df=pd.read_csv("https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/dataset_part_1.csv")
df.head(10)
```

```
[3]:
```

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Long
0	1	2010-06-04	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003	-80.5'
1	2	2012-05-22	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0005	-80.5'
2	3	2013-03-01	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0007	-80.5'
3	4	2013-09-29	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	0	B1003	-120.6'
4	5	2013-12-03	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1004	-80.5'
5	6	2014-01-06	Falcon 9	3325.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1005	-80.5'
6	7	2014-04-18	Falcon 9	2296.000000	ISS	CCAFS SLC 40	True Ocean	1	False	False	True	NaN	1.0	0	B1006	-80.5'
7	8	2014-07-14	Falcon 9	1316.000000	LEO	CCAFS SLC 40	True Ocean	1	False	False	True	NaN	1.0	0	B1007	-80.5'
8	9	2014-08-05	Falcon 9	4535.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1008	-80.5'
9	10	2014-09-07	Falcon 9	4428.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1011	-80.5'

Identify and calculate the percentage of the missing values in each attribute

```
[4]: df.isnull().sum()/len(df)*100
```

```
[4]: FlightNumber      0.000000
Date                0.000000
BoosterVersion      0.000000
PayloadMass         0.000000
Orbit               0.000000
LaunchSite          0.000000
Outcome             0.000000
Flights             0.000000
GridFins            0.000000
Reused              0.000000
Legs                0.000000
LandingPad          28.888889
Block               0.000000
ReusedCount         0.000000
Serial              0.000000
Longitude           0.000000
Latitude            0.000000
dtype: float64
```

Identify which columns are numerical and categorical:

```
[5]: df.dtypes
```

```
[5]: FlightNumber      int64
Date                object
BoosterVersion      object
PayloadMass         float64
Orbit               object
LaunchSite          object
Outcome             object
Flights             int64
GridFins            bool
Reused              bool
Legs                bool
LandingPad          object
Block               float64
ReusedCount         int64
Serial              object
```

TASK 2: Calculate the number and occurrence of each orbit

Use the method `.value_counts()` to determine the number and occurrence of each orbit in the column `Orbit`

Note: Do not count GTO, as it is a transfer orbit and not itself geostationary.

```
[6]: # Apply value_counts on Orbit column
df['Orbit'].value_counts()
```

```
[6]: Orbit
GTO      27
ISS      21
VLEO     14
PO        9
LEO        7
SSO        5
MEO        3
ES-L1     1
HEO        1
SO         1
GEO        1
Name: count, dtype: int64
```

TASK 3: Calculate the number and occurrence of mission outcome of the orbits

Use the method `.value_counts()` on the column `Outcome` to determine the number of `landing_outcomes`. Then assign it to a variable `landing_outcomes`.

```
[7]: # landing_outcomes = values on Outcome column
landing_outcomes = df['Outcome'].value_counts()
landing_outcomes
```

```
[7]: Outcome
True ASDS      41
None None      19
True RTLS      14
False ASDS      6
True Ocean      5
False Ocean      2
None ASDS        2
False RTLS        1
Name: count, dtype: int64
```

```
None ASDS      2  
False RTLS      1  
Name: count, dtype: int64
```

`True Ocean` means the mission outcome was successfully landed to a specific region of the ocean while `False Ocean` means the mission outcome was unsuccessfully landed to a specific region of the ocean. `True RTLS` means the mission outcome was successfully landed to a ground pad `False RTLS` means the mission outcome was unsuccessfully landed to a ground pad. `True ASDS` means the mission outcome was successfully landed to a drone ship `False ASDS` means the mission outcome was unsuccessfully landed to a drone ship. `None ASDS` and `None None` these represent a failure to land.

```
[ ]: for i,outcome in enumerate(landing_outcomes.keys()):  
      print(i,outcome)
```

We create a set of outcomes where the second stage did not land successfully:

```
[ ]: bad_outcomes=set(landing_outcomes.keys()[[1,3,5,6,7]])  
bad_outcomes
```

TASK 4: Create a landing outcome label from Outcome column

Using the `Outcome`, create a list where the element is zero if the corresponding row in `Outcome` is in the set `bad_outcome`; otherwise, it's one. Then assign it to the variable `landing_class`:

```
[ ]: # landing_class = 0 if bad_outcome  
      # landing_class = 1 otherwise
```

This variable will represent the classification variable that represents the outcome of each launch. If the value is zero, the first stage did not land successfully; one means the first stage landed Successfully

```
[ ]: df['Class']=landing_class  
df[['Class']].head(8)
```

```
[ ]: df.head(5)
```

We can use the following line of code to determine the success rate:

```
[ ]: df["Class"].mean()
```

We can now export it to a CSV for the next section, but to make the answers consistent, in the next lab we will provide data in a pre-selected date range.

```
df.to_csv("dataset_part 2.csv", index=False)
```